

Prospective Acceptance and Use of Artificial Intelligence-Based Voice, Speech, and Facial Expression Features in Cancer Patients' Journey

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Abstract

Emotional distress is often present throughout a cancer patient's journey. Although screening is essential in cancer management, several barriers hinder implementation, such as care providers' workload, cumbersome paper-and-pencil or digital surveying, or inaccurate monitoring of distress trajectories. Evidence shows psychosocial interventions significantly reduce distress and improve quality of life once detected. However, early detection and access to these resources face barriers like long waiting times and restrictions on patients' time schedules, work, mobility, or economic status, preventing commutes to healthcare centers. Multimodal alternatives using Artificial Intelligence (AI) can detect distress by analyzing voice, facial expressions, and speech content, which can work in ubiquitous devices and are effortless for patients. This project aims to research cancer patients' prospective acceptance and use of these AI-powered tools throughout the phases of cancer management.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); HCI theory, concepts and models; Human computer interaction (HCI); Empirical studies in HCI; • **Applied computing** → Life and medical sciences; Health informatics.

Keywords

acceptance, mental health, perception, psycho-oncology

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1 Introduction

Emotional distress is a significant burden for cancer patients and survivors, impacting their overall well-being [1, 2]. Untreated distress can lead to poorer quality of life, treatment adherence, and

medical outcomes [3]. Consequently, the importance of distress screening has been recognized as an essential component of comprehensive cancer care [4, 5].

Implementing a successful screening for emotional distress has several barriers [6]. First, healthcare providers often lack the time or sufficient expertise to accurately assess and intervene in cases of psychological distress [7]. Second, psychological distress surveys (either paper-and-pencil or digital) are often cumbersome to administer during a patient's visit, and many patients find completing the same survey multiple times repetitive and taxing [8, 9]. Finally, even in cancer centers that have successfully implemented distress screening protocols, assessments tend to be one-time efforts and do not monitor the trajectory of psychological distress longitudinally. This cross-sectional approach is problematic because cancer patients' distress trajectories fluctuate over time as they encounter different clinical phases—from the diagnosis and coping with treatment to survivorship and the fear of recurrence [6, 10]. These distinct phases trigger fluctuating psychological states that a single screening event can miss.

The failure to detect distress fluctuations is critical because effective treatments are available. Meta-analyses have demonstrated that psychological interventions (e.g. Cognitive Behavioral Therapy, Mindfulness-based therapies) significantly reduce distress and improve quality of life in cancer populations [11, 12]. However, the efficacy of these interventions is nullified if the patient's distress is never identified. Consequently, when screening fails or occurs too late, patients lose access to these evidence-based support systems.

Recent advances in artificial intelligence (AI) based distress assessment tools focusing on language, speech, or facial expressions are being tested to address the above challenges [13, 14]. Spontaneously generated cues to psychological states and their recognition by AI can lead to improved screening for emotional distress [13]. More specifically, trimodal approaches (i.e., language, speech, and facial expression) have the potential to improve the accuracy of predictive models and reduce the likelihood of misclassifying signs of emotional distress [14].

As with any emerging technology, the effective adoption of AI-based voice, speech semantics, and facial expression (AIVSFE) tools requires a fundamental understanding of users' willingness to accept and use them [15, 16]. In this context, acceptance implies the utilization of an intervention, as expressed in uptake rates, adherence, or satisfaction [17]. However, cancer patients' and survivors' prospective perception (i.e., future expectations), acceptance, and utilization of AIVSFE tools remain largely unexplored [13].

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2 Project Methodology

2.1 Research Questions

Given the novelty of AIVSFE tools applied to cancer-related emotional distress, this research would contribute to answering the following questions:

- What are the facilitators and barriers that could drive cancer patients to accept and use AIVSFE tools?
- Would acceptance and use vary along the different clinical phases (active treatment, survivorship, and long-term survivorship)?
- Do sociodemographic or clinical factors impact the acceptance and use of AIVSFE tools?
- Are the acceptance and use of the three AIVSFE tools different?
- Are there differences in acceptance and use among cancer patients who are currently experiencing emotional distress and those who are not?

2.2 Study Objectives

General Objective:

1. Explore cancer patients' prospective acceptance and use of AIVSFE tools to inform the ethical and responsible development of clinical implementation.

Specific Objectives:

1. Review the current state of AI multimodal perception and usability studies among cancer patients and survivors.
2. Understand the perceptions and experiences of cancer patients regarding AI and its application in cancer management solutions (qualitative approach).
3. Study the prospective acceptance and use variations between AIVSFE tools (quantitative approach).
4. Study the differences in prospective acceptance and use across the different stages of cancer (quantitative approach).
5. Assess the impact of sociodemographic and clinical factors on the prospective acceptance and use of AIVSFE tools (quantitative approach).
6. Evaluate differences in the perception of AIVSFE tools between cancer patients with and without current emotional distress (quantitative approach).

2.3 Study Design

This project has been structured into three phases, which are based on an exploratory sequential design [18]. Phase 1 entailed a scoping review of AI multimodal perception and usability studies administered to cancer patients and survivors. Phase 2 utilized a qualitative approach to gather in-depth, exploratory insights into the experiences, perceptions, and concerns of cancer patients and survivors regarding AI in cancer management. Lastly, Phase 3 will utilize a structured survey to validate and generalize the themes identified in Phase 2, thereby facilitating a broader understanding of cancer patients' and survivors' acceptance of AI-based tools. Further aspects of each phase are detailed next.

3 Phase 1 & 2 Contributions

3.1 Cancer Patients' Perception, Acceptance, and Utilization of Artificial Intelligence-Based Emotional Distress Assessment Tools: A Scoping Review

A scoping review was conducted to synthesize, for the first time, the available evidence on perception, acceptance, and utilization of AIVSFE tools for detecting emotional distress in cancer patients and survivors [19]. This scoping literature review was conducted according to the PRISMA Extension for Scoping Reviews (PRISMA-ScR) guidelines [20] and was prospectively registered in PROSPERO (CRD42024509595).

Inclusion criteria were empirical studies that (a) reported quantitative or qualitative data, published between 2019 and 2025; (b) involved adult (≥ 18 years) cancer patients at any stage of treatment or survivorship; (c) described their perception, acceptance, or utilization of AIVSFE screening tools for emotional distress; and (d) were published in English or Spanish. Exclusion criteria were (a) studies involving child or adolescent cancer patients or survivors, and (b) palliative care studies at end-of-life phase. The focus on the adult population stems from the understanding that this demographic group exhibits distinct characteristics and needs that differ from those of other younger age groups or advanced clinical stages. Moreover, the search of documents since 2019 is due to our interest in the most recent evidence in this novel topic.

A total of 1,598 abstracts were identified from six databases and screened against our inclusion criteria. From this pool, 21 records were retrieved from the database and registry searches, and 5 records were retrieved via citation searching. Three articles were included in the final review [21–23]. The publication dates for the included articles ranged from 2022 to 2024.

3.1.1 Speech Semantics Methodologies and User Feedback. The Multilingual Framework for Risk Assessment and Symptom Tracking (MRAST) [22] and the Artificial Intelligence-based Co-Facilitator (AICF) [21] are innovative approaches to harnessing speech data to improve patient care. The MRAST prompts patients to record video diaries that capture their symptoms and experiences in real time. The framework utilizes an automatic speech recognition system to analyze the video recordings and extract symptoms and relevant information from the patients' spoken words, enabling a detailed understanding of their health status. Additionally, natural language processing (NLP) techniques are used to process the extracted data and identify symptoms and possible causes. An ad-hoc Likert scale from 1 (poor) to 10 (excellent) was used to collect user feedback about the MRAST at three points in time (i.e., at the start of the app-based questionnaire, after the introduction of the virtual agent, and at the end of the study). Overall usability received an average rating of 76%, their diary recording experience 70%, and their experience with structured questionnaires 75%.

The AICF was developed to identify keywords that indicate psychosocial concerns and to recommend appropriate online resources to members of an online support group (OSG) who express such a concern. The model was trained using a corpus of chat sessions from the CancerChatCanada (CCC) OSG and feedback from human annotations, providing language and expression resources used by

cancer patients. The AICF utilizes an NLP model to create vector representations of terms and phrases related to psychosocial concerns expressed by participants. Once these concerns have been identified, the AICF uses a matrix to match these concerns with appropriate online resources, taking into consideration the patient's specific attributes. An ad-hoc binary scale ("useful" and "not useful") was used to collect participant feedback on this approach. Of the participants who accessed the resources recommended by the AICF, 75% rated them as "useful." Therefore, although only two studies were considered, results suggest that AI-based tools analyzing speech semantics are considered useful by most participants, leading to positive experiences.

3.1.2 Facial Expression Methodology and User Feedback. The Sophisticated High-speed Object Recognition Engine (SHORE) [23] software employs an object recognition engine to obtain patient data through real-time emotion recognition. SHORE analyzes facial expressions based on underlying action units (AUs), which are then classified into basic facial emotions such as "happy," "angry," "sad," "surprised," and "neutral." The software utilizes a camera-based video stream to capture images of individuals in a separate room within the oncology outpatient clinic where participants have been invited to perform the experiment. An ad-hoc questionnaire on meaningfulness and stressfulness was used to gather feedback from users: 98% of participants reported that they did not find the real-time emotion analysis to be stressful, and 88% considered the real-time analysis of their emotional state while waiting to be meaningful. No significant differences in acceptance were reported according to age or gender.

The perception, acceptance, and usability outcomes of these studies were positive, ranging from 70% to 98%. Despite the limited number of studies that met the inclusion criteria, highlighting their early-stage development, the findings underscore promising advances in the use of AIVSFE tools to assess and address patient needs. Altogether, the participants in these studies responded positively, suggesting that these technologies are potentially well perceived by patients in the context of cancer care. However, the present state of AIVSFE tools has not yet surmounted the inherent challenges that preclude their comparison with current emotional distress tools for oncology populations. Despite the potential benefits of these tools, including their increased ubiquity and reduced effort, further evidence from more robust study designs and replicable outcome measures is crucial to encourage their adoption by healthcare providers, users, and policymakers.

3.2 Cancer Patients and Survivors' Perspectives on Artificial Intelligence Tools for Emotional Distress Detection: A Qualitative Study

After reviewing the literature on the perception of AIVSFE tools, a qualitative study was conducted using the Unified Theory of Acceptance and Use of Technology (UTAUT) as an implementation framework. The study aimed to understand the perspectives of cancer patients' and survivors' perspectives on the potential facilitators and barriers of AIVSFE tools in distress detection. This model has

been empirically tested in multiple contexts to explain the factors that underlie an individual's decision to adopt technology [24, 25].

Twenty adult patients and survivors diagnosed with any cancer type were recruited for this study. Semi-structured interviews were audio-recorded, transcribed verbatim, and analyzed to identify determinants, barriers, and facilitators. Thematic analysis was conducted including transcription, familiarization, initial coding, theme searching, reviewing themes, defining and naming themes, and producing the report [17]. The constructs of UTAUT model provided a deductive approach for the initial phase of analysis, which were iteratively revised throughout the entire process.

Preliminary results identified four and three themes within the perceived facilitators and barriers, respectively. The perceived facilitators reported were (1) the recommendation and supervision from a healthcare professional and endorsement of competent authorities, (2) immediate accessibility without delay and immediate responses, (3) neutrality, free from human biases and accurate clinical memory, and (4) precise detection and support for early diagnosis. Barriers identified were (1) dehumanization and a deficiency in empathy, (2) mistrust of data vulnerability and undisclosed interests, and (3) inability to manage emotional subjectivity and provide tailored recommendations. Overall, most participants expressed their intention to use AIVSFE tools.

These results from cancer patients and survivors provide insights for prospective adoption of AIVSFE tools in psycho-oncological settings. Incorporating the identified facilitators and barriers into implementation strategies, including responsible and ethical AI policies, will help to ensure effective integration of AIVSFE tools into emotional distress assessments. However, larger-scale studies are needed to determine which predictors would facilitate the implementation of these tools into clinical practice.

4 Phase 3 protocol

In consideration of the contributions from phases 1 and 2, the final phase will enable the measurement of key constructs across a larger and more representative sample. This will yield statistically robust conclusions that will complement the in-depth insights from the qualitative phase.

4.1 Study Design

The hypothesized technology adoption model can be seen in Figure 1. Extended e-health constructs (i.e., trust, perceived risks, hedonic motivation) have been added to the four core constructs of the UTAUT model.

A survey will be designed and divided into four sections:

1. Sociodemographic and clinical questionnaire: This ad-hoc section will collect sociodemographic (e.g., age, gender, education) and clinical data (e.g., type of cancer, stage, current treatments).
2. AI Perception Survey: This section will assess cancer patients' acceptance and potential usage based on the proposed UTAUT model (Figure 1).
3. Distress Visual Analog Scale [26]: Participants will be asked to complete a single-item visual analog scale ranging from 0 to 10, with the aim of assessing their own emotional distress during the previous week.

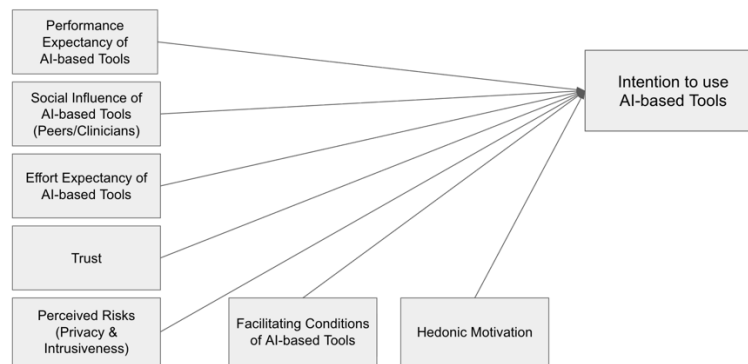


Figure 1: Research model with UTAUT core and extended predictors of AIVSFE tools intention to use.

4. E-health Literacy: Participants will be asked to complete the eHealth Literacy Scale (eHEALS) [27] to assess their digital literacy.

4.2 Procedure

A small group of volunteers will assess the survey and item comprehension and usability before its implementation. Based on previous research on technology acceptance [28], an anticipated large effect size ($f^2 = 0.35$) and aiming for a conventional power level of 0.80 and a significance level of 0.05, a sample size of 480 participants will provide sufficient statistical power for reliable parameter estimation in the SEM model. This decision is supported by the rule of thumb recommending 10 participants per parameter when large effect sizes are expected [29]. After the participant has been briefed and has accepted, the online questionnaires will be facilitated for completion. Quantitative data will be collected through the GDPR-compliant software Qualtrics for online surveying.

Preliminary statistical analysis will be executed (descriptive and normality analysis), and validity and reliability will be tested for self-developed scales for the survey. Differences between the three modalities of AI tools, clinically distressed and non-distressed patients according to the Distress Visual Analog Scale, and sociodemographic groups will be assessed with mean and proportion differences as appropriate (e.g., Student's *t*, Kruskal-Wallis, chi-squared). The Structural Equation Modelling (SEM) technique will be utilized to analyze statistical correlations between the dependent and independent variables collected through the survey. Analyses will be conducted using SPSS v24.0.

Recruitment of participants will be held within healthcare institutions, cancer patient associations, and social media. Inclusion criteria consist of being 18 years or older; being diagnosed with any type of cancer irrespective of the time elapsed since treatment completion, cancer location or stage; being able to write and read Spanish; and having Internet access. Exclusion criteria consist of the inability to comprehend instructions and questions, undergoing end-of-life care, or having current severe psychiatric comorbidities.

4.3 Ethical Implications

This research involves the participation of humans, who will be informed of their rights before they decide to join the study. Participants will be informed that 1) participation is voluntary, 2) no compensation is foreseen, and 3) their refusal to participate or opt out at a later stage will have no consequences whatsoever. Given the non-physical nature of interventions, no physical harm is anticipated, but any adverse event will be reported. All procedures comply with the ethical principles of the UNESCO Universal Declaration of Bioethics, the Helsinki Declaration, and the Belmont Report. European GDPR (EC/2016/679) and the Spanish Personal Data Protection Law (LOPD 3/2018), along with its associated methods to securely collect, store, access, and share data have been devised, will be followed to ensure the confidentiality of all collected data. Data will not be transferred to third countries or parties. The project has been approved by the leading institution Ethics Committee.

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